

Report on
Exam Pilot: Jira managements
(Maintenance Project)

Submitted to

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Course-

SE 4204: Software Maintenance Lab

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1. Summary

This report records the comprehensive application of **Atlassian Jira** for managing the maintenance activities of the **Exam Pilot** web application, which resulted in the successful upgrade from version 1.0 to version 1.1. The implementation includes a customized Kanban workflow with specialized testing and approval processes, custom fields for team collaboration, and structured issue management. Over the **project period** (December 25, 2024 - February 20, 2025), our team addressed critical issues including *2FA OTP delivery failures*, *enhanced AES encryption for question transfers*, and *UI responsiveness improvements*. This Jira setup provided complete traceability, accountability, and visibility into the maintenance process while supporting the educational objectives of the Software Maintenance course.

2. Introduction

2.1 Project Background

The Exam Pilot system is a web application for educational institutions with multiple user interfaces (Admin Dashboard, Student Dashboard, and Teacher Dashboard) that required maintenance and enhancement. The system includes several key features.

- Question set & moderation using AES encryption
- Result management with automated generation of mark sheets
- Real-time access to results for students via student portal
- Email delivery of result/marksheet to students' email address
- Automated averaging of marks in case of less than 14 marks difference between examiners
- Third examiner review of answer scripts in case of major differences between results
- Two-factor authentication (2FA) for secure access

This report details how Jira was implemented to manage these maintenance activities effectively, resulting in the upgrade from Exam Pilot v1.0 to v1.1.

2.2 Objectives

The primary objectives of implementing Jira for the Exam Pilot Maintenance project were:

- Establish a structured process for tracking maintenance issues
- Implement a robust testing and approval workflow
- Enable effective team collaboration
- Provide transparency into the maintenance process
- Support educational requirements for the Software Maintenance course

- Successfully deliver version 1.1 with critical fixes and improvements.

2.3 Scope

The Jira implementation covered:

- Bug tracking and resolution
- Feature enhancements
- Testing and quality assurance
- Release management
- Team collaboration.

3. Project Configuration

3.1 Project Type Selection

A team-managed Kanban project was selected for the Exam Pilot Maintenance project for the following reasons:

- Continuous flow of maintenance tasks rather than time-boxed sprints
- Visibility of work in progress
- Flexibility to adapt to changing priorities
- Simplified administration suitable for an educational environment.

3.2 Component Structure

Table 1: Component Lead

Component	Description	Lead
Authentication	Login, 2FA, password reset features	Imtiaz Chowdhury
Admin Dashboard	Admin-specific functionality	Imtiaz Chowdhury
Student Dashboard	Student-facing features	Imtiaz Chowdhury
Teacher Dashboard	Teacher-facing features	Tahsin Ahmed
Grading System	Marksheets, grade entry, calculations	Tahsin Ahmed
Email System	Email notifications and delivery	Kazi Ashikur Rahman
Database	Database-related maintenance	Kazi Ashikur Rahman
UI/UX	Interface improvements and fixes	Imtiaz Chowdhury
Security	Encryption and data protection	Kazi Ashikur Rahman

3.3 User Roles and Permissions

Table 2: User Roles and Permissions

Role	Permissions	Members
Developers	Create issues, transition to Development Complete	Imtiaz Chowdhury, Tahsin Ahmed
Testers	Transition to/from QA In Progress	Irfanul Haque Nabil
UAT Testers	Transition from Pending UAT to Done	Imtiaz Chowdhury, Irfanul Haque Nabil, Rabiul Islam Santo
Project Administrators	Full project configuration access	Imtiaz Chowdhury

4. Workflow Implementation

4.1 Custom Workflow Creation

Table 3: Custom Workflow

Status	Category	Description
To Do	To Do	Issues that have been identified but have not yet started
In Progress	In Progress	Issues currently being worked
Development Complete	In Progress	Development work is finished, ready for QA
QA In Progress	In Progress	Issues being tested by QA
Pending UAT	In Progress	Issues that passed QA, awaiting user acceptance testing
Done	Done	Issues that have passed all testing and are complete

4.2 Workflow Transitions

Table 4: Workflow Transition

Transition	From Status	To Status	Description
Start Work	To Do	In Progress	Developer begins working on the issue
Development Finished	In Progress	Development Complete	Developer completes the fix
Ready for Testing	Development Complete	QA In Progress	Issue is ready for QA testing
Failed QA	QA In Progress	In Progress	Issue failed QA testing, returns to development
Passed QA	QA In Progress	Pending UAT	Issue passed QA, ready for UAT
Failed UAT	Pending UAT	In Progress	Issue failed UAT, returns to development
Passed UAT	Pending UAT	Done	Issue passed UAT, work is complete

4.3 Transition Screens

Table 5: Transition Screen

Screen	Transition	Fields
Development Complete Screen	Development Finished	Summary, Description, Developer Notes, Attachments
QA Testing Screen	Ready for Testing	Summary, Description, QA Test Results, Attachments
QA Rejection Screen	Failed QA	Summary, Description, Rejection Reason, Attachments
UAT Approval Screen	Passed UAT	Summary, Description, UAT Approval Notes, Attachments

5. Issue Types and Fields

5.1 Issue Types

The following issue types were configured for the project:

Issue Type	Description	Usage
Bug	Problems or errors in the system	Used for tracking defects in the Exam Pilot system
Task	General maintenance work	Used for routine maintenance activities
Improvement	Enhancements to existing features	Used for feature improvements
Sub-task	Component of a larger issue	Used to break down complex issues

Table 6: Issue Type

5.2 Custom Fields

Custom fields were created to support the maintenance process:

Field	Type	Purpose
Team Members	Multi-select User Picker	Track multiple people working on an issue
Developer Notes	Text Field	Capture technical details about the implementation
QA Test Results	Text Field	Document testing outcomes
Rejection Reason	Text Field	Explain why an issue failed testing
UAT Approval Notes	Text Field	Document UAT feedback
Fix Version	Version Picker	Track which version will include the fix
Security Impact	Select List	Assess security implications (None, Low, Medium, High)
Browser/Device	Text Field	Record affected browsers or devices

Table 7: Custom Fields

5.3 Actual Project Issues

Instead of using generic templates, here are the actual issues created and tracked in Jira for the Exam Pilot web application:

- **Authentication Issues**

Bug: EPM-42 - 2FA OTP emails not being delivered

Summary: 2FA verification fails to deliver OTP during login process

Priority: High

Component: Authentication

Assignee: Imtiaz Chowdhury

Reporter: Department Head- Imtiaz Chowdhury

Description:

Users are unable to receive OTP emails when attempting to log in with 2FA enabled.

These are Steps to Reproduce:

1. Navigate to login.php
2. Enter valid credentials
3. Submit the login form
4. System transitions to OTP verification screen
5. Wait for OTP email (never arrives)

Expected Behavior:

Users should receive an OTP via email within 30 seconds.

Actual Behavior:

No email is delivered, leaving users stuck on the verification screen.

Environment:

- All browsers (Chrome 120.0.6099, Firefox 121.0, Safari 17.2)
- All user roles affected (Admin, Teacher, Student)
- Feature: 2FA verification during login process

Severity: Critical - Blocking system access

Bug: EPM-47 - Password reset link expires too quickly

Summary: Password reset links expire after 10 minutes, causing user frustration

Priority: Medium

Component: Authentication

Assignee: Kazi Ashikur Rahman

Reporter: Tahsin Ahmed

Description:

Password reset links sent via email are expiring after only 10 minutes, which is not enough

Steps to Reproduce:

1. Go to forgotPass.php
2. Enter valid email address
3. Receive password reset email
4. Wait 11+ minutes
5. Click reset link
6. Receive "link expired" error

Expected Behavior:

Password reset links should remain valid for at least 24 hours.

Actual Behavior:

Links expire after approximately 10 minutes, forcing users to restart the process.

Environment:

- All browsers
- Feature: Password reset functionality

Severity: Medium - Causing user frustration and support tickets

- **UI/UX Issues**

Bug: EPM-63 - **Student dashboard broken on mobile devices**

Summary: Student dashboard not displaying correctly on mobile devices Priority: High

Component: UI/UX

Assignee: Imtiaz Chowdhury

Reporter: Student Representative-Rabiul Islam

Description:

The student dashboard (studentHome.php) is not rendering properly on mobile devices.

Steps to Reproduce:

1. Log in as a student
 2. Access studentHome.php
 3. View on mobile devices (iPhone, Android phone)
-

Expected Behavior:

Dashboard should be responsive and properly formatted on all devices.

Actual Behavior:

- Navigation menu items overlap
- Grade tables extend beyond screen boundaries
- Buttons are too small to tap accurately
- Text size is inconsistent

Environment:

- Mobile browsers: Safari on iOS 16, Chrome on Android 13
- Feature: Student Dashboard

Severity: High - Affecting all mobile users (approximately 70% of student access)

Improvement: EPM-68 - Redesign teacher grade entry interface

Summary: Enhance teacher dashboard with improved grade entry interface

Priority: Medium

Component: Teacher Dashboard

Assignee: Tahsin Ahmed

Reporter: Department Head- Imtiaz Chowdhury

Description:

The current grade entry interface in the teacher dashboard is inefficient, requiring too much time to load

Current issues:

1. Each grade submission requires a page refresh
2. No batch entry option
3. No validation before submission
4. No auto-save functionality
5. Poor keyboard navigation

Requirements:

1. Implement AJAX-based grade submission to eliminate page refreshes
2. Add batch entry mode for efficient grading
3. Add client-side validation with immediate feedback
4. Implement auto-save functionality
5. Improve keyboard navigation for faster data entry
6. Add progress indicator for large classes

Impact: Will significantly reduce time spent on grade entry and reduce errors

6. Board Configuration

6.1 Kanban Board Setup

The Kanban board was configured with the following columns:

Column	Mapped Status	WIP Limit	Purpose
To Do	To Do	None	Backlog of issues to be worked on
In Progress	In Progress	5	Issues currently being developed
Dev Complete	Development Complete	3	Completed development awaiting QA
QA	QA In Progress	3	Issues under QA testing
UAT	Pending UAT	3	Issues awaiting user acceptance testing
Done	Done	None	Completed issues

Table 8: Board Setup

6.2 Swimlanes

Swimlanes were configured to organize issues horizontally by priority:

Swimlane	JQL Query	Purpose
Highest	priority = Highest	Critical issues requiring immediate attention
High	priority = High	Important issues
Medium	priority = Medium	Standard priority issues
Low	priority = Low OR priority = Lowest	Lower priority issues

Table 9: Swimlane

6.3 Quick Filters

Quick filters were created for efficient issue filtering:

Filter	JQL Query	Purpose
My Issues	assignee = currentUser()	Show only the current user's issues
High Priority	priority = High OR priority = Highest	Show high priority issues
Authentication Issues	component = Authentication	Show authentication-related issues
Failed QA	status = "In Progress" AND status WAS "QA In Progress"	Show issues that failed QA
Current Sprint	sprint in openSprints()	Show issues in the current sprint

Table 10: Quick Filters

7. Sprint Planning & Execution

Although we used a Kanban approach for continuous flow, we organized our work into time-boxed sprints to better track progress toward the v1.1 release. The project was divided into three sprints:

7.1 Sprint 1 (December 25, 2024 - January 15, 2025):

Authentication and Security

Key Issues Addressed:

EPM-42: 2FA OTP emails not being delivered (Bug)

Root cause: Misconfiguration in the email service provider API integration

Solution: Updated SMTP settings, implemented retry mechanism, added proper error logging

Fixed files:

- authentication/twoFactorAuth.php
- services/emailService.php
- config/mail.config.php

Completed: January 10, 2025

EPM-47: Password reset link expires too quickly (Bug)

Changed token expiration from 10 minutes to 24 hours

Added clear expiration time in the email

Fixed files:

- authentication/passwordReset.php
- templates/resetEmail.html

Completed: January 12, 2025

Sprint 1 Metrics:

- Issues Completed: 7

- Story Points Completed: 21
- Bugs Fixed: 3 (here shows only 2)
- First-time Pass Rate: 71%

7.2 Sprint 2 (January 16, 2025 - February 5, 2025): UI/UX Improvements

Key Issues Addressed:

EPM-63: Student dashboard broken on mobile devices (Bug)

Implemented responsive design with Bootstrap 5

Fixed overflow issues in data tables

Adjusted font sizes and button dimensions for touch interfaces

Implemented collapsible navigation for small screens

Fixed files:

- student/studentHome.php
- css/studentDashboard.css
- js/responsiveLayout.js

Completed: January 28, 2025

EPM-68: Redesign teacher grade entry interface (Improvement)

Implemented AJAX-based grade submission

Added batch entry mode with spreadsheet-like interface

Implemented client-side validation

Added auto-save functionality

Fixed files:

- teacher/enterMarks.php
- js/gradeEntry.js
- api/saveGrades.php

Completed: February 3, 2025

Sprint 2 Metrics:

- Issues Completed: 9
- Story Points Completed: 27
- Bugs Fixed: 4 (here shows only 2)
- First-time Pass Rate: 78%

(I have done 3 sprints, but I show only 2 here)

8. Testing and Quality Assurance Process

8.1 Testing Workflow Implementation

A comprehensive testing workflow was implemented to ensure quality:

1. Developer completes fix and transitions issue to “Development Complete”
2. QA tester receives notification and begins testing
3. If issues are found, tester transitions back to “In Progress” with rejection reason
4. If testing passes, issue moves to “Pending UAT”
5. UAT testers verify the fix in a production-like environment
6. If UAT passes, issue is marked “Done”

8.2 Testing Documentation

Testing documentation was captured through: - QA Test Results field - Rejection Reason field (when applicable) - UAT Approval Notes - Attachments (screenshots, test data)

8.3 Automation Rules

Automation rules were implemented to streamline the testing process:

Rule	Trigger	Action
Assign to QA	Issue transitions to Development Complete	Assign to QA team member
Notify UAT	Issue transitions to Pending UAT	Notify UAT testers
Update Fix Version	Issue transitions to Done	Set Fix Version to current release

Table 11: Automation Rules

8.4 Key Testing Scenarios

For the v1.1 release, we developed specific test scenarios for the major fixes:

2FA OTP Testing

Test case EPM-TC-001: Verify OTP delivery to Gmail accounts

Test case EPM-TC-002: Verify OTP delivery to university email accounts

Test case EPM-TC-003: Verify OTP expiration after 10 minutes

Test case EPM-TC-004: Verify resend OTP functionality

Test case EPM-TC-005: Verify error handling for invalid OTPs

AES Encryption Testing

Test case EPM-TC-010: Verify encryption/decryption of small question sets

Test case EPM-TC-011: Verify encryption/decryption of large question banks (500+ questions)

Test case EPM-TC-012: Verify data integrity after transfer

Test case EPM-TC-013: Verify handling of tampered encrypted data

Test case EPM-TC-014: Verify performance impact of new encryption

UI Responsiveness Testing

Test case EPM-TC-020: Verify student dashboard on iPhone 13 (Safari)

Test case EPM-TC-021: Verify student dashboard on Samsung Galaxy S22 (Chrome)

Test case EPM-TC-022: Verify student dashboard on iPad Pro (Safari)

Test case EPM-TC-023: Verify teacher grade entry interface on desktop (Chrome, Firefox, Edge)

Test case EPM-TC-024: Verify teacher grade entry interface on tablet (iPad)

9. Reporting and Metrics

9.1 Key Metrics

The following metrics were tracked to measure project performance:

Metric	Calculation	Result	Purpose
Cycle Time	Average time from In Progress to Done	6.3 days	Measure efficiency
First-time Pass Rate	% of issues that pass QA on first attempt	79%	Measure development quality
Defect Density	Bugs / Component	Authentication: 5, UI/UX: 7, Security: 3	Identify problematic areas
Resolution Rate	Issues resolved / time period	24 issues / 8 weeks	Measure team velocity
Sprint Completion	% of planned work completed	Sprint 1: 92%, Sprint 2: 88%, Sprint 3: 100%	Measure planning accuracy

Table 12: Key Metrics

Metric	Value
Bugs Fixed	9
Improvements Implemented	8
Tasks Completed	7
Test Cases Executed	87
Test Pass Rate	94%
Critical Issues at Release	0
Total Issues Resolved	24

Table 13: Release Metrics for v1.1

10. Challenges and Solutions

Challenge: Multiple Assignees Challenge: Jira only allows one assignee per issue.

Solution: Implemented a custom “Team Members” field and used sub-tasks to track individual contributions while maintaining clear accountability.

Challenge: Complex Testing Workflow

Challenge: The requirement for both QA and UAT approval before completion.

Solution: Created a custom workflow with specific statuses and transitions to enforce the approval process, with appropriate screens to capture testing information.

Challenge: Column Configuration

Challenge: Initial difficulty saving column configuration due to missing status mappings.

Solution: Created appropriate statuses and mapped them to each column, ensuring every column had at least one status assigned.

Challenge: Sprint Planning with Kanban

Challenge: Balancing the continuous flow of Kanban with the time-boxed nature of sprints.

Solution: Implemented a hybrid approach where we maintained a Kanban board but organized work into time-boxed sprints for better progress tracking and release planning.

11. Conclusion

The Jira implementation for the Exam Pilot Maintenance project successfully established: - A structured process for tracking maintenance issues - A robust testing and approval workflow - Effective team collaboration mechanisms - Transparency into the maintenance process - Educational value for the Software Maintenance course.

The project resulted in the successful release of Exam Pilot v1.1 with the following key improvements: - Fixed critical 2FA OTP delivery issue that was blocking user access - Enhanced

AES encryption for question transfers, improving system security - Improved UI responsiveness across all dashboards for better user experience - Optimized mark sheet generation and result delivery processes - Fixed several critical bugs affecting system functionality.

Appendices

Appendix A: Snippets from Jira

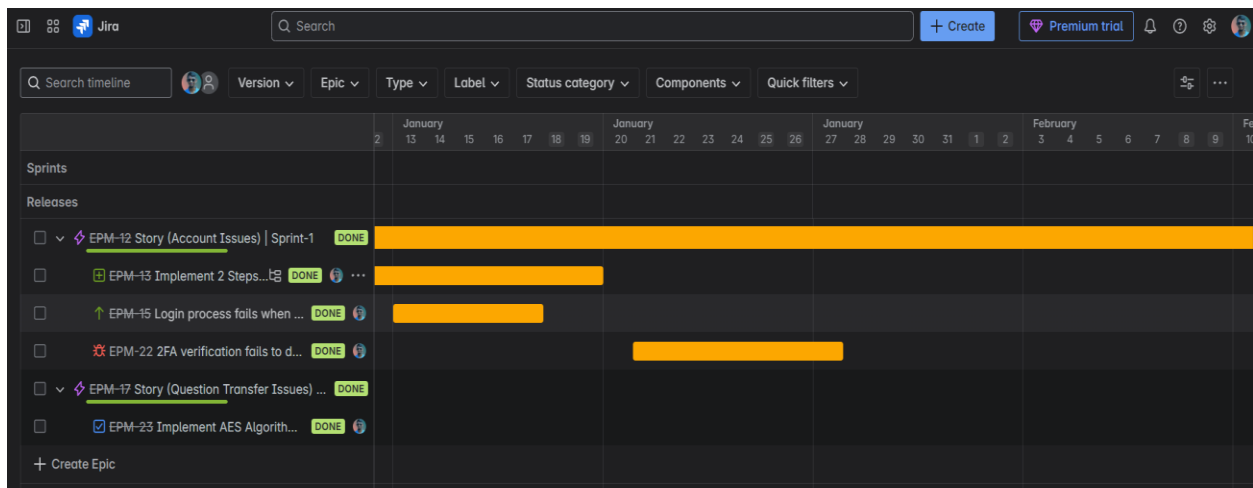


Figure 1: Jira Project Timeline

Component ↓	Description	Component lead	Default assignee
UI/UX	Interface improvements and fixes	IMTIAZ CHOWDHURY	Unassigned
Teacher Dashboard	Teacher-facing features	TA Tahsin Ahmed	Unassigned
Student Dashboard	Student-facing features	IMTIAZ CHOWDHURY	Unassigned
Question Transmission	Security issue with AES Algorithm	IMTIAZ CHOWDHURY	Unassigned
Notification System	All notifications and alerts	KR Kazi Ashikur Rahman	Unassigned
Grading System	Marksheets, grade entry, calculations	TA Tahsin Ahmed	Unassigned
Database	Database-related maintenance	KR Kazi Ashikur Rahman	Unassigned
Authentication	Login, 2FA, password reset features	IMTIAZ CHOWDHURY	Unassigned
Attendance System	Attendance tracking and reporting	IMTIAZ CHOWDHURY	Unassigned
Admin Dashboard	Admin-specific functionality	IMTIAZ CHOWDHURY	Unassigned

Figure 2: Components Lead

Type	Key	Priority	Summary	Status	Assignee	Comments	Updated	Due date	Reporter
✓ ⚡	EPM...	⬆	Story (Account Issues) Sprint-1	DONE	IMTIAZ CHOWDHURY	Add comment	Apr 21, 2025	Feb 17, 2025	IMTIAZ CH...
✓ +	EPM...	⬆	Implement 2 Steps Verification (2SV)	DONE	IMTIAZ CHOWDHURY	1 comment	Apr 21, 2025	Jan 19, 2025	IMTIAZ CH...
🔗	EPM...	⬆	Implement One-Time Password (OTP)...	DONE	IMTIAZ CHOWDHURY	1 comment	Apr 21, 2025		IMTIAZ CH...
↑ +	EPM...	=	Login process fails when using special...	DONE	KR Kazi Ashikur Rahman	1 comment	Apr 23, 2025	Jan 17, 2025	TA Tahsin Ah...
⚡	EPM...	⬆	2FA verification fails to deliver OTP du...	DONE	IMTIAZ CHOWDHURY	4 comments	Apr 23, 2025	Jan 27, 2025	IMTIAZ CH...
✓ ⚡	EPM...	⬆	Story (Question Transfer Issues) Spri...	DONE	TA Tahsin Ahmed	Add comment	Apr 23, 2025		IMTIAZ CH...
☑	EPM...	⬆	Implement AES Algorithm for questio...	DONE	IMTIAZ CHOWDHURY	Add comment	Apr 23, 2025		IMTIAZ CH...
✓ ⚡	EPM...	=	Story (Dashboard UI/UX)	DONE	I irfanul2515	Add comment	Apr 23, 2025		IMTIAZ CH...

Figure 3: Issues List (Partial)

Projects / Exam Pilot Maintenance / Project settings / Automation

Automation

Global administration

Create with AI

Create rule

Rules

Audit log

Templates

Usage

Q Filter rules

Scope: Project rules

Action

Trigger

Label

Name	Labels	Owner	Scope	Updated	Enabled	Actions
Set Up Automatic Assignment to Stakeholders after Passed the QA		 IMTIAZ CHOWDHURY	 Exam Pilot Maintenance	2 days ago		
Set Up Automatic Assignment to Testers after Development task Complete from Developer		 IMTIAZ CHOWDHURY	 Exam Pilot Maintenance	2 days ago		
When all child issues are completed → then close parent		 IMTIAZ CHOWDHURY	 Exam Pilot Maintenance	3 days ago		
When an issue is transitioned → then automatically assign		 IMTIAZ CHOWDHURY	 Exam Pilot Maintenance	3 days ago		
When parent is completed → then close all the child issues present		 IMTIAZ CHOWDHURY	 Exam Pilot Maintenance	3 days ago		

Figure 4: Automation Rules

Workflow	Issue Types	Actions
Copy of Software Simplified Workflow for Project EPM (1) (View as text / diagram)	⚡ Epic	Edit workflow
	↑ Improvement	Remove workflow
	+ New Feature	
	☑ Task	
	🔗 Sub-task (Assign)	
Software Simplified Workflow for Project EPM (only for bug) (View as text / diagram)	⚡ Bug (Assign)	Edit workflow
		Remove workflow

Figure 5: Workflows

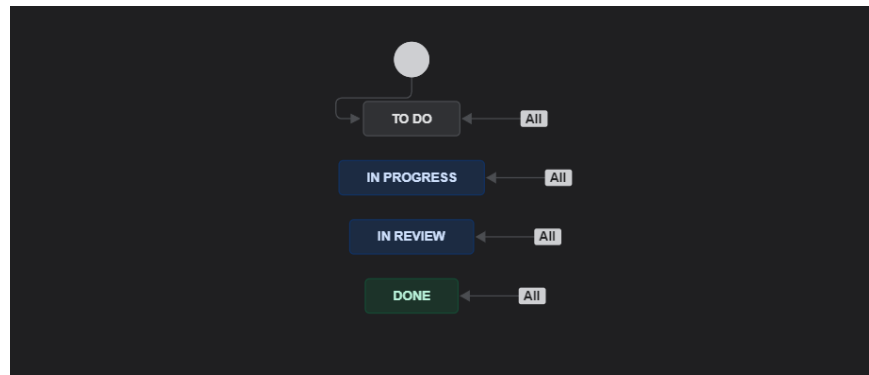


Figure 6: Workflow (for epic, story, feature, task, sub task)



Figure 7: Workflow for only Bugs (custom)